Specification Document

Meal Planner App for the Elderly

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1. Introduction

1.1 Purpose of the Document

This Specification Document explains exactly what the **Meal Planner App for the Elderly** should do and how it should behave for users. It shows the functional requirements through use cases and diagrams, includes GUI mock-ups, and lists testable non-functional requirements.

The document acts as a clear guide for development and testing. It helps make sure the final app matches what users expect and meets the project goals set out in the Vision Document. It connects the high-level ideas from research to practical implementation details that can actually be tested.

1.2 Scope of the System

This section demonstrate what the app will and will not do. Setting clear boundaries helps keep the project focused and achievable, especially within the given time period.

In scope for this specification:

- User profile creation including basic personal details, dietary preferences, allergies, health considerations.
- Personalized weekly meal plan generation based on user's profile, with manual user selection of meals.
- Shopping list creation from selected meals.
- Basic barcode scanning support to assist with grocery items for shopping.
- Simple meal tracking to record completed meals.

Out of scope for this specification:

- Advanced user authentication and account management.
- Payment processing.
- Real-time collaboration between users over multiple devices.

2. System Overview

2.1 Brief Description of the Meal Planner App

The Meal Planner App for the Elderly is a web application that helps older users create weekly meal plans and manage shopping more easily. It takes the user's profile information - like dietary preferences, allergies, and health conditions - and uses this to show suitable meal options from a pre-approved list.

Users start by setting up a simple profile with their basic details and food requirements. The app then lets them pick meals for each day of the week (breakfast, lunch, dinner) from options that match their needs. Once the weekly plan is complete, the system automatically creates a shopping list with all the ingredients needed.

The app includes basic barcode scanning so users can check grocery items against their shopping list or add new items quickly. There's also simple meal tracking where users can mark meals as completed to see their progress through the week.

Everything runs through a web browser on phones, tablets, or computers. No app download is needed. The design uses large buttons, clear text, and high contrast colours following accessibility guidelines [1]. For the prototype, meal data comes from local sample files rather than external services.

Caregivers can help by using the same device to view the shopping list and mark items as bought. The focus stays on making healthy eating simple for elderly users while reducing the daily planning effort.

3. Actors / User Roles

3.1 Elderly User

The Elderly user is the main user who uses the Meal Planner App. Typically aged 65 or over, they want to eat healthy but find the process of meal planning difficult and hard. They might have specific dietary neds, allergens, or health conditions that limited food choices.

In the app, their journey is simple:

Set up profile including meal preference(health conditions, allergens, dietary needs, budget).

Pick their meals for the week from approved meals the app just created for them.

Use the app, follow the shopping list and scan items while shopping and check items from shopping list.

They mainly use app on their own devices like mobile, tablet, laptop thorough a web browser. The interface should be simple and clean with large text and buttons as the app's main users are elderly peoples.

3.2 Caregiver / Family Member

The caregiver or family member helps elderly user with meal selection, preparation and in shopping. This might any person who helps elderly user. The caregiver will create own account then add elderly using UUID for assistance. They might also look at meal plan to know what food is needed or help the elderly person in meal preparation and also check the shopping list.

4. Functional Requirements

4.1 User Profile Management

Before getting any meal plan the user needs to set up their profile. The app lets user to create and update profile that stores information required for meal planning. After login to app first time, they see a profile setup screen where they can enter some personal and dietary information which is required.

Profile information includes:

- Personal details e.g. age group, name, height(optional)
- Dietary selection e.g. vegetarian, gluten-free, dairy-free nuts-free, etc.
- Health considerations e.g. hyper-tension, diabetes, high blood pressure, etc.

Once the information is saved, it stays in the app until the user changes it. The system will then uses this information to filter and recommend meals that fits with the user preference.

4.2 Meal Planning and Recommendation

The app shows meal options that matches with the user's preference and let users to create weekly meal plan including breakfast, lunch, dinner for each day.

How meal planning work step-by-step:

- After setting up profile, the meal options grouped by type e.g. breakfast, lunch, dinner will display to user
- Only approved and exact matching meals will be appear.
- Each meal options shows name, short description, dietary name e.g. vegetarian, dairy-free, etc.

Meal selection process step-by-step:

- The system guides users through the weekly meal planning process one day at a time.
- For each day, users select meals for breakfast, lunch, and dinner.
- Users may replace or change selected meals before moving to the next day.
- The system provides clear progress feedback to indicate how many meals have been selected for the week.
- Users can review and modify the full weekly plan before finalising it.

4.4 Shopping List Generation

The app support the creation of shopping list based on the meals plan. This feature will reduce the effort involved in grocery planning for elderly users and caregivers.

- Shopping list will be automatically generated using ingredients from the selected meals in the weekly plan.
- Duplicate ingredients from different meals should be combine into a single item entry.
- Shopping list will be downloadable.
- Users can review the shopping list after the weekly meal plan has been completed.
- Shopping list can be update when changes are made to the selected meals.

Caregivers or family members may assist with viewing and managing the shopping list using shared access to the application.

4.5 Barcode Scanning Functionality

The app provides basic barcode scanning functionality to assist users during grocery shopping. This feature will reduce the effort required to manage shopping items and to minimise mistakes such as forgetting or duplicating items. The app allow:

- Users to scan the barcode of grocery items using the device camera.
- Scanned items to be checked against the existing shopping list.
- Scanned items to be added to the shopping list if they are not already present.

Barcode scanning is provided as a convenience feature to support shopping tasks and does not require integration with external retail systems, pricing information, or delivery services [4].

5. Use Case Diagram

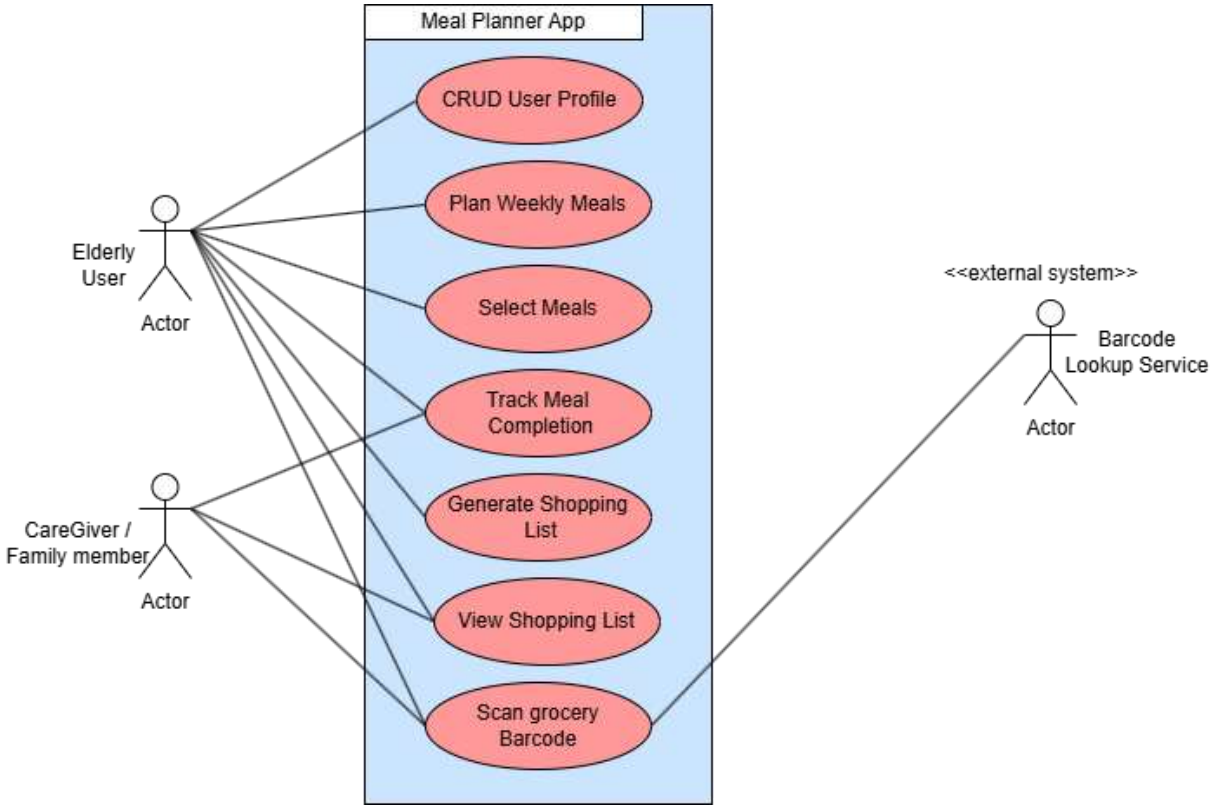

Fig.1. Use case diagram [Author's own]

6. Use Case Descriptions

6.1 Generate Weekly Meal Plan

Use Case : Generate Weekly Meal Plan

Primary Actor: Elderly user

Secondary Actor: Caregiver

Description:

This use case describes how a user create meal plan by selecting meals for each day based on their preferences.

Preconditions:

1) The user has created profile for meal planning.
2) The user has access to meal planning feature.

Main Flow:

1) User opens the meal planning screen showing 7 days grid Monday-to-Sunday with breakfast, lunch, dinner.
2) Mondays slots are active, rest of the slots greyed out/un-clickable.
3) User click on Monday.
4) App displays filtered meals for Monday.
5) All 3 meals selected, Tuesday slots available.
6) User continuous until all the 7 days meal selected.
7) After selecting all 7 day meals, **Create Shopping list** button becomes active.

Alternate Flow:

1) The user can change or replace previously selected meal before completing meal plan.

Postconditions:

1) Meal plan is created and stored.
2) Plan is ready for shopping list generation.

6.2 Select a Meal

Use Case : Select a Meal

Primary Actor: Elderly user

Secondary Actor: caregiver / Family Member

Description:

This use case describes how can a user select meals and meal type for specific day while weekly meal planning.

Preconditions:

1) The user started weekly meal planning.
2) Suitable meal options are available for the user based on their preference.

Main Flow:

1) The user selects a specific day within the weekly meal planning.
2) The app displays meal for each breakfast, lunch, dinner.
3) The user select meals from displayed meals.
4) The app assigned the meal to the chosen day.
5) The app update the weekly meal plan to show selection.

Alternate Flow:

1) The user update or change the selected meal.
2) The app update the meal with new meal selection.

Postconditions:

1) The selected meal is saved as part of the weekly meal plan.

6.3 Scan Grocery Barcode

Use Case: Scan Grocery Barcode

Primary Actor: Elderly User

Secondary Actor: Caregiver / Family Member

External system: Barcode lookup Service

Description:

This use case describes how a user can scans a grocery item barcode to manage shopping list during shopping. The app will provide immediate feedback showing whether the scanned item are on the list or not.

6.4 View Shopping List

Use Case: View Shopping List

Primary Actor: Elderly User

Secondary Actor: Caregiver / Family Member

Description:

This use case describes how a user can view shopping list which is generated from the selected meals of weekly meal plan. The shopping list will display all the items which is need in a clear, structured and organized manner. Elderly users and caregivers both can use shopping list to support meal preparation and shopping task.

6.5 Track Meal Completion

Use Case: Track Meal Completion

Primary Actor: Elderly User

Secondary Actor: Caregiver / Family Member

Description:

This use case describes how a use can records their meal completion during the week. By checked meals as completed, the app allow users and caregivers / family member to track progress against meal plan.

7. GUI Mock-ups / Wireframes

7.1 Home / Dashboard Screen

This screen provides the main entry point to the application in mobile and web browser.

Fig.2. Home Screen, Mobile interfaces [Author's own]

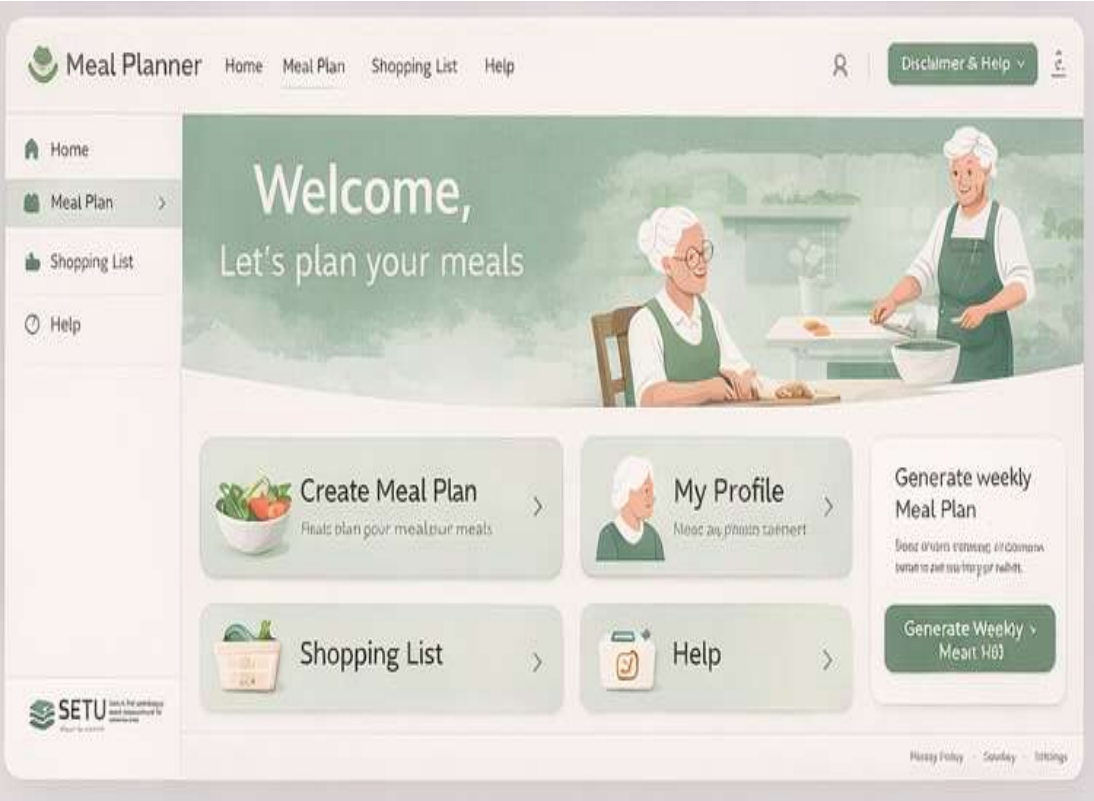

Fig.3. Home Screen, Web-interfaces [Author's own]

7.2 Meal Plan Screen

This screen supports the weekly meal planning process by guiding users through daily meal selection.

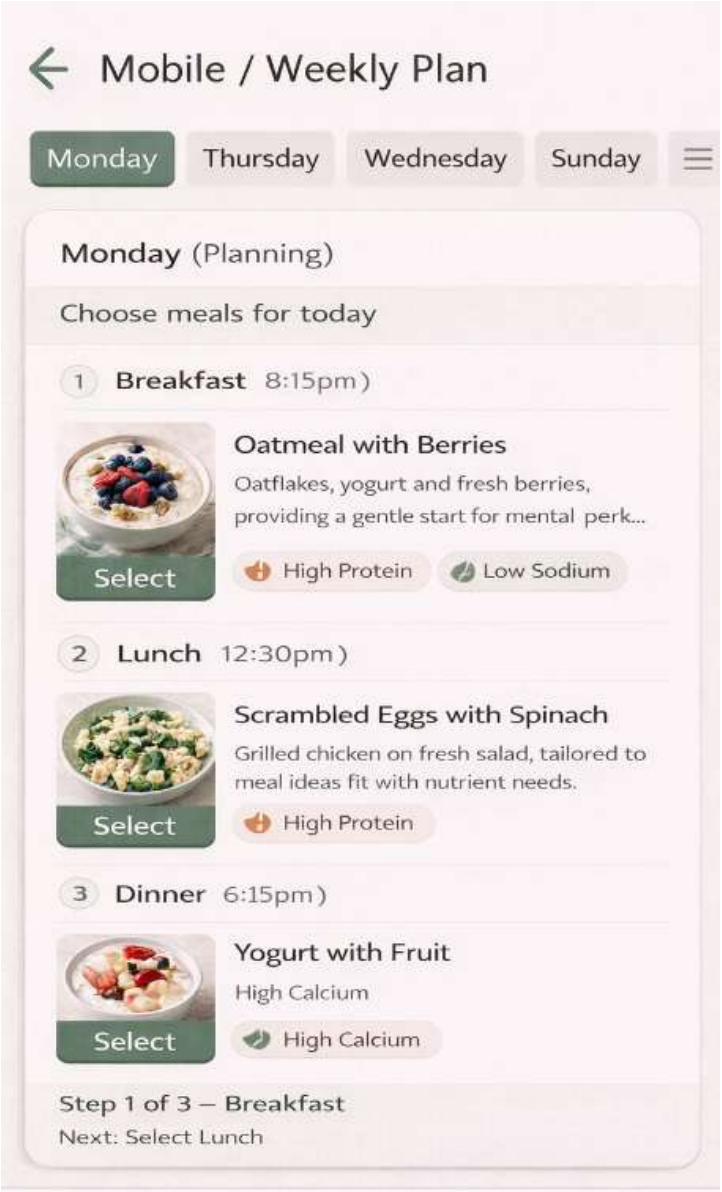

Fig.4. Meal planning Screen, Mobile-interfaces [Author's own]

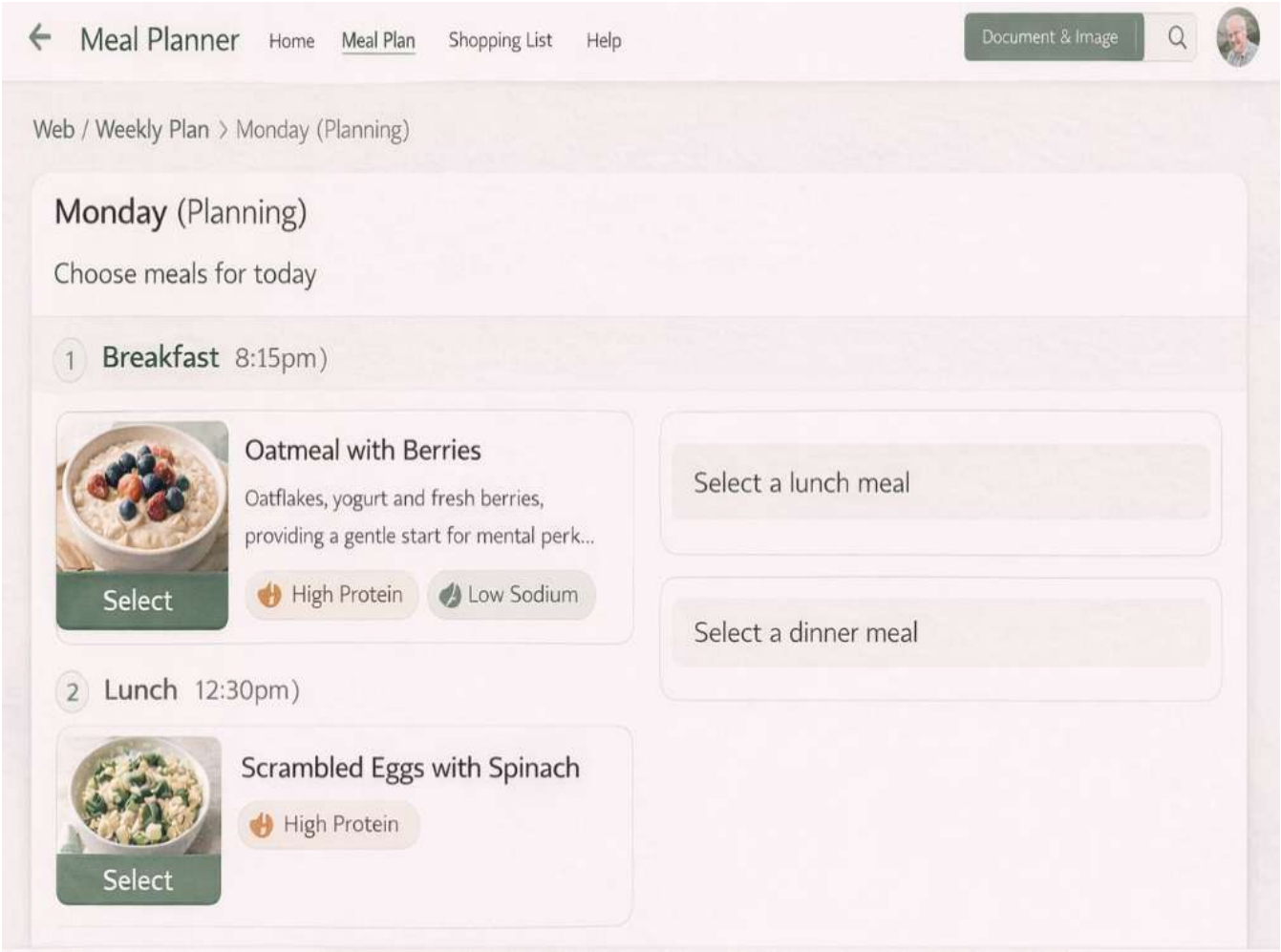

Fig.5. Meal planning Screen, Web-interfaces [Author's own]

7.3 Meal Details Screen

This screen displays detailed information about a selected meal to support informed selection.

Fig.6. Meal Description Screen[Author's own]

7.4 Shopping List Screen

This screen presents the shopping list generated from the weekly meal plan to assist users during grocery shopping.

Fig.7. Shopping List Screen [Author's own]

7.5 Barcode Scanning Screen

This screen allows users to scan grocery item barcodes to assist with shopping list management.

Fig.8. Barcode Scanning Screen [Author's own]

7. Non-Functional Requirements

8.1 Usability Requirements

1) **Simple profile setup**

The system should allow new users to complete profile setup in a short and reasonable time period.

Test Method: Elderly users creating an initial profile.

Acceptance Criteria: Average completion time must be less than 120 seconds.

2) **Efficient Meal Selection**

The system shall allow users to select meals efficiently.

Test Method: Users select meals for a single day within time specified time period.

Acceptance Criteria: Average completion time must be less than 90 seconds.

These usability requirements are informed by established usability principles for elder adults [3].

8.2 Accessibility Requirements

1) **Text Readability and Contrast**

The system should display text at a size that supports readability for elderly users.

Test Method: Font sizes inspected using browser developer tools, Use a contrast checking tool.

Acceptance Criteria: Body text is at least 18px and headings are at least 24px, All text meets WCAG 2.1 AA contrast requirements minimum 4.5:1 [2].

2) **Touch Target Size**

The system should ensure that all interactive elements are suitable for touch based interactions.

Test Method: Buttons dimensions measured using browser developer tools

Acceptance Criteria: All primary buttons have a minimum size of 48 × 48 pixels

8.3 Reliability Requirements

1) **Data Persistence**

The system should retain user profile data within application session.

Test Method: Open application, save profile, close application/browser, reopen

application, and verify stored data

Acceptance Criteria: User data stored correctly and safely after reopening application.

2) **Error Handling for Invalid Input**

The system should handle invalid or unreadable barcode scans without crashing application.

Test Method: Attempt to scan an invalid barcode or user enter text instead of numbers in input field.

Acceptance Criteria: The system displays an appropriate error message and continues to function normally.

8.4 Platform Constraints

1) **Web Based Platform**

The system should be accessible through modern web browsers.

Test Method: Access application using modern web browser.

Acceptance Criteria: The application load and work correctly without need of installation.

2) **Device Compatibility**

The system support common devices such as mobiles, tablets, laptops, and desktop computers.

Test Method: Performed responsive testing on all devices.

Acceptance Criteria: The core functionality of system remains useable and accessible across all devices.

9. Conclusion

The functional and non-functional requirements of the Meal Planner App for the Elderly have been clearly and systematically described in this specification document. The document outlines the system's scope, user roles, core features, and quality requirements in order to support effective meal planning and shopping assistance for elderly users and their caregivers.

The document explains how users are expected to interact with the system through functional requirements, a use case diagram, detailed use case descriptions, and GUI mock-ups. Provides a strong foundation for the design, development, and testing of the system.

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[4] Dix et al., *Human–Computer Interaction*, 3rd ed., Pearson Education, 2004. (**Accessed**: 03 February 2026)

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